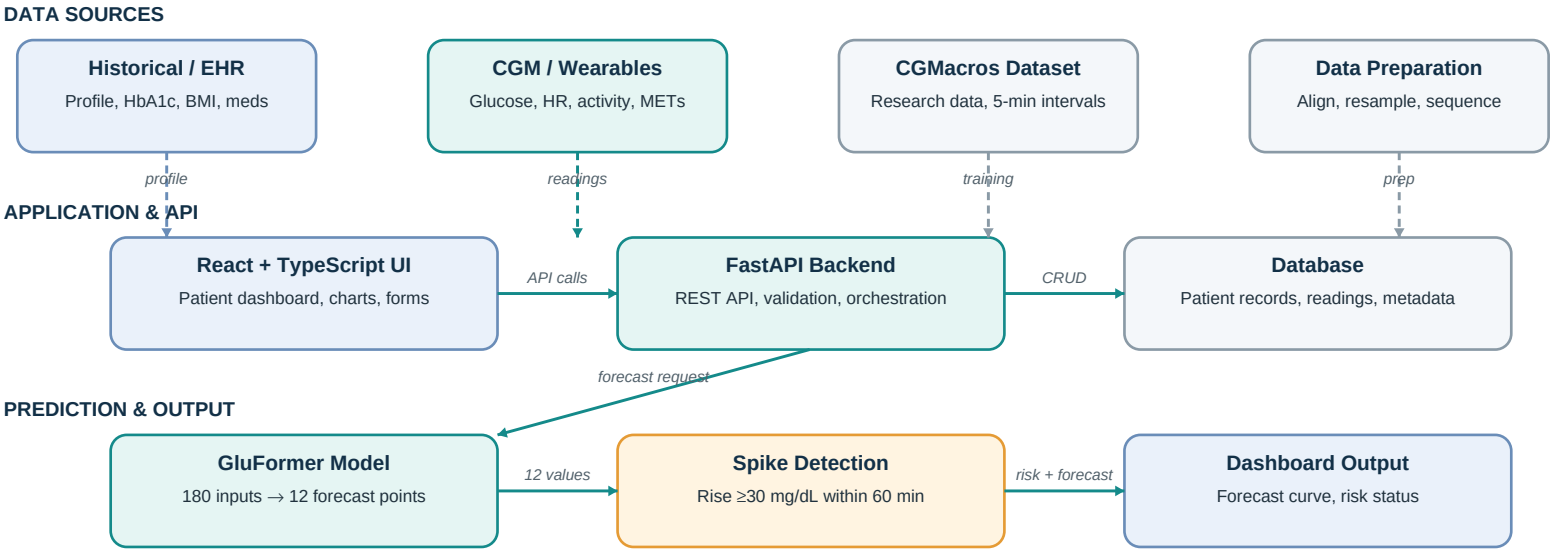
Diabetes Digital Twin — System Architecture

GluFormer-based 60-minute glucose forecasting | Happiest Health Digital Twin Challenge 2026



Conceptual design. The current web prototype does not yet have verified GluFormer inference connected.

1. Collect	2. Prepare	3. Serve	4. Forecast	5. Display
Patient profile and CGM / wearable readings	Align timestamps, resample to 5-minute intervals, and form a 180-point input sequence	React UI calls FastAPI; patient data and readings are stored in the database	Compatible GluFormer inference should return 12 points for 60 minutes; flag a rise ≥30 mg/dL	Display forecast curve and spike-risk status on the clinician-facing dashboard

Important limitation: The React + TypeScript frontend and FastAPI workflow are represented as the prototype architecture, but verified GluFormer inference is not yet wired into the web app. Research metrics were measured on six held-out CGMacros participants and are not clinical validation.